

CS489 – Natural Language Processing
Eugene Baek

Project Proposal
20820288

1. What data will you use?

The project is specific to space mission communication. By analyzing the existing data in ``data/open-dev/input.txt``, the sentences appear to be transcripts from NASA missions. For example, in line 1 of `input.txt`, Lake Champlain refers to the aircraft carrier used to recover “the first crewed Project Mercury mission (Freedom 7)...” ([https://en.wikipedia.org/wiki/USS_Lake_Champlain_\(CV-39\)](https://en.wikipedia.org/wiki/USS_Lake_Champlain_(CV-39)))).

To replicate this domain, I will be using well-formatted transcripts from space missions made available at <https://github.com/Spacelog/Spacelog/tree/main>.

2. What model will you consider?

The model I will use is Google’s CANINE model. CANINE fits the project specification since it is a character-level model; unlike other models (e.g. BERT), CANINE uses Unicode characters. This allows CANINE to support multilingual inputs and out-of-vocab characters as it uses hash-based embeddings that can represent any Unicode character.

Using the pretrained CANINE model, we will fine-tune the model using LoRA trainable layers on the curated space mission dataset. These will generate latent embeddings which will feed into a feed-forward network (FFN) as a classification task on our vocabulary. The top-3 predictions will be submitted.

We note that CANINE is a bidirectional model and not an autoregressive model. However, we adapt the model to predict next characters by using a causal mask on the inputs and a FFN for classification. This is appropriate as this is not strictly an autoregressive task (i.e. single next character vs next words) and autoregressive language models often do not offer the character fidelity we require (i.e. they use subwords/BPE).

3. How will you design your system?

We first build the dataset from the transcripts. This involves preprocessing the data: normalize whitespaces, remove tags or identifiers, split paragraphs into newline-delimited sentences. As an additional step, we will translate the sentences across all specified languages to increase data variety and prevent class imbalance.

From the dataset, we can (1) create a vocabulary to be used by the FFN and (2) generate training and validation data as a next token prediction problem (e.g. input=hello wo; output=r).

As a baseline, we create the CANINE model without LoRA layers and only train the FFN on the dataset. We then create the CANINE model with LoRA layers and train both LoRA layers and FFN layers on the dataset. Measuring both the baseline and the fine-tuned model will give us a good idea on the performance gain from domain-specific fine-tuning.

There remains undecided design choice (e.g. layer dimensions, latent embedding dimension, vocab construction details, activation functions, hyperparameters) that will be answered throughout experimentation.